

Milan Air Pollution and Respiratory Morbidity

Structure and period

Item	Value
Total observations	155 weekly ISO records
Period	2023-W18 → 2026-W16
Raw columns	33 (daily pollutant values + outcomes + meteo)
Final columns (analysis_ready.csv)	39 (after adding identifiers, flags, harmonics)
Missing values — outcomes	0 (respiratory_disease, ILI, pneumonia)
Missing values — exposures	0 (NO2_mean, PM25_mean, PM10_mean)
Missing values — meteo	0 (temp_mean, humidity_mean)
Partial exposure weeks	0 — all 155 weeks have 7 valid pollutant days
W52/2025 flag	1 week flagged (partial; included in main analysis)
Outlier W8/2025	1 week flagged (anomalously low counts; excluded from seasonal analysis)
Zero-count ILI weeks	0 — no structural zeros

Outlier: Week 8/2025

Variable	Value W8/2025	Winter mean	Assessment
Respiratory disease	1,095 visits	3,075 visits	↓ -64% vs winter mean
ILI	438 visits	1,087 visits	↓ -60% vs winter mean
Pneumonia	224 visits	439 visits	↓ -49% vs winter mean

Decision: W8/2025 excluded from seasonal analysis (Steps 5–6) but retained in analysis_ready.csv with outlier flag = 1 for DLNM sensitivity analysis.

Descriptive Statistics (n = 155 weeks)

Table 1. Full descriptive statistics for all variables.

Variable	N	Mean	Median	SD	Min	P25	P75	Max
POLLUTANTS								
NO ₂ mean (µg/m³)	155	33.12	31.21	12.01	12.82	22.93	42.02	66.14
PM _{2.5} mean (µg/m³)	155	19.58	14.69	12.35	5.74	11.21	24.62	72.91
PM ₁₀ mean (µg/m³)	155	29.07	24.43	14.78	11.27	19.44	34.51	94.29
METEOROLOGY								

Temperature (°C)	155	14.54	14.64	7.72	1.05	7.90	20.88	30.35
Humidity (%)	155	76.92	78.62	11.02	31.97	69.96	85.54	94.93
HEALTH OUTCOMES								
Resp. disease (visits/wk)	155	2647	2656	693	1095	2289	2886	4940
ILI (visits/wk)	155	932	882	326	438	722	985	2122
Pneumonia (visits/wk)	155	373	358	135	144	276	429	763

Seasonal comparison (winter vs summer, outlier excluded)

Table 2. Mann-Whitney U comparison: winter (n=83) vs summer (n=39).

Variable	Winter mean	Winter SD	Summer mean	Summer SD	p (Mann-Whitney)
Resp. disease	3074.5	580.3	1956.3	362.5	p < 0.001 ***
ILI	1086.5	358.3	690.7	125.3	p < 0.001 ***
Pneumonia	439.3	134.1	282.9	96.8	p < 0.001 ***
NO ₂ (µg/m ³)	41.5	9.4	23.7	7.0	p < 0.001 ***
PM _{2.5} (µg/m ³)	26.0	12.8	12.0	3.8	p < 0.001 ***
PM ₁₀ (µg/m ³)	36.3	15.5	21.1	6.1	p < 0.001 ***
Temperature (°C)	8.7	4.4	22.9	3.5	p < 0.001 ***
Humidity (%)	81.5	11.0	71.5	7.3	p < 0.001 ***

Spearman Correlation & Collinearity

Key Spearman correlations (n = 154, outlier excluded)

Table 3. Pollutant–outcome and inter-pollutant Spearman correlations.

Pair	Spearman ρ	Interpretation	Implication
Temperature → Resp.	−0.827	Very strong inverse	⚠ Dominant confounder
NO ₂ → Resp.	+0.665	Strong positive	★ Primary association
NO ₂ → ILI	+0.500	Moderate positive	★ Significant
NO ₂ → Pneumonia	+0.456	Moderate positive	★ Significant
PM _{2.5} → Resp.	+0.531	Moderate positive	★ Significant
PM _{2.5} → ILI	+0.453	Moderate positive	✓ Significant
PM _{2.5} → Pneumonia	+0.404	Moderate positive	✓ Significant
PM ₁₀ → Resp.	+0.496	Moderate positive	★ Significant

PM₁₀ → ILI	+0.449	Moderate positive	✓ Significant
PM₁₀ → Pneumonia	+0.445	Moderate positive	✓ Significant
NO₂ ↔ PM_{2.5}	+0.802	High	⚠ Collinear — single-pollutant models
NO₂ ↔ PM₁₀	+0.765	High	⚠ Collinear
PM_{2.5} ↔ PM₁₀	+0.956	Near-perfect	⚠ Very high — do NOT combine

Collinearity conclusion (communicated to P2): PM_{2.5} and PM₁₀ have $\rho = 0.956$. All DLNM models are single-pollutant (9 models: 3 pollutants × 3 outcomes).

Confounding conclusion (communicated to P2): Temperature ($\rho = -0.827$ with Resp.) is controlled with a natural cubic spline (df=3) as a within-season confounder — SEPARATELY from the Fourier seasonal component.

EDA Figures

All figures produced at 300 dpi. Season colours: Winter = blue, Spring = green, Summer = orange-red, Autumn = orange. [Figure 1 — Weekly mean pollutant time series](#)

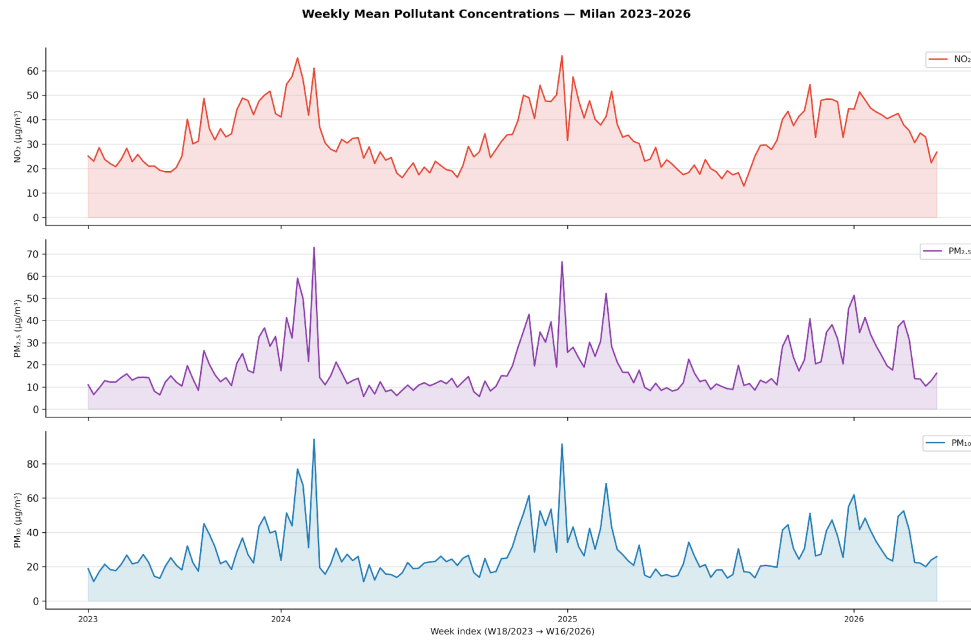


Figure 1. Weekly mean NO₂, PM_{2.5} and PM₁₀ concentrations — Milan W18/2023–W16/2026 (n=155). Clear winter peaks visible for all three pollutants.

Figure 2 — Health outcome time series

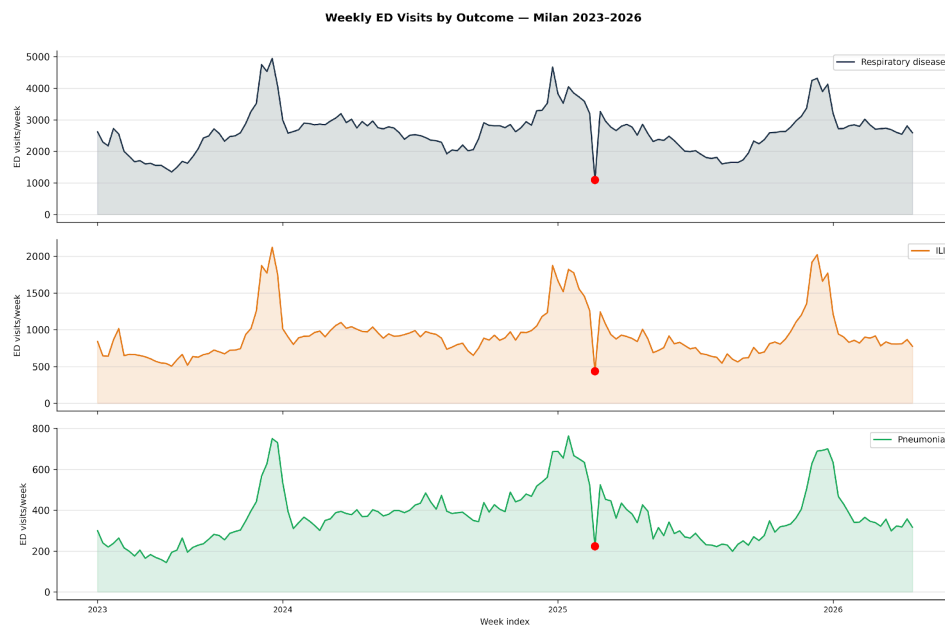


Figure 2. Weekly ED visits for respiratory disease, ILI and pneumonia. Red dot = outlier W8/2025. Strong winter seasonality visible in all outcomes.

Figure 3 — Standardised pollutant concentrations (Z-scores)

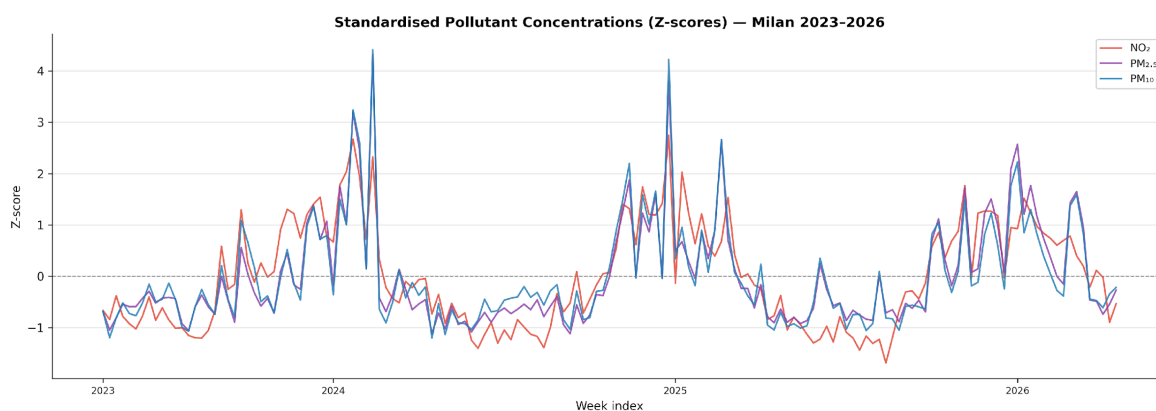


Figure 3. Z-score normalised concentrations of NO₂, PM_{2.5} and PM₁₀ on a shared scale. Confirms synchronised seasonal behaviour and high inter-pollutant collinearity.

Figure 4 — Meteorological variables

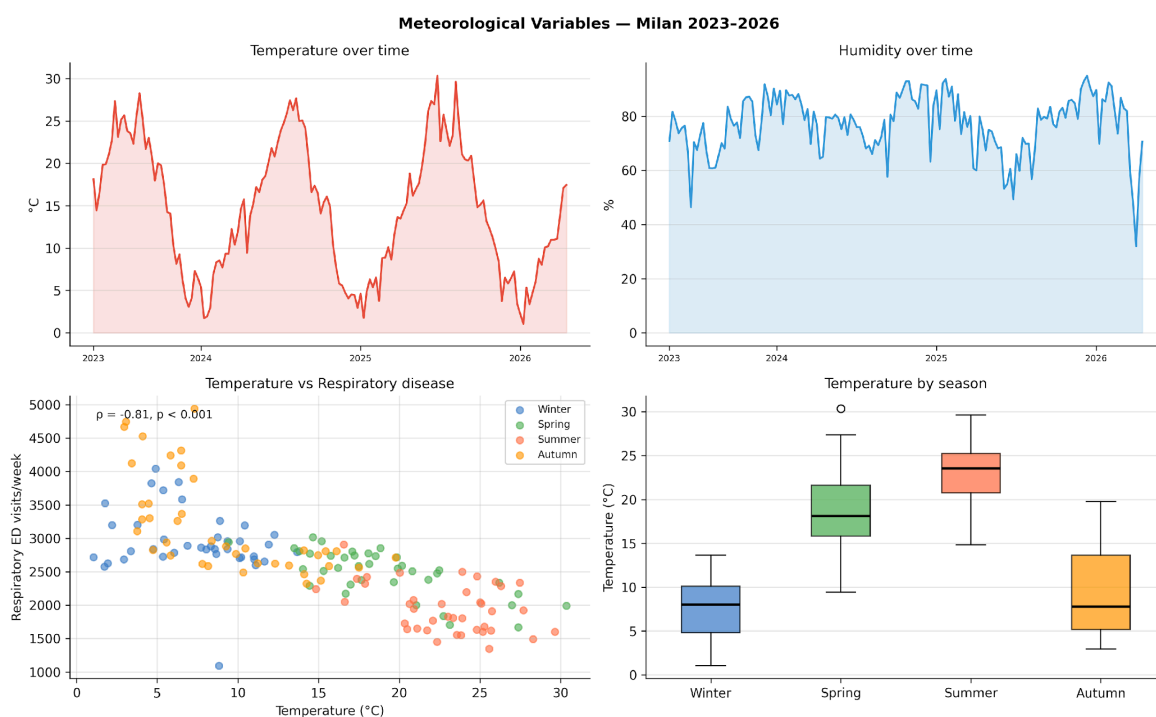


Figure 4. Top: temperature and humidity over time. Bottom left: temperature vs respiratory disease scatter ($p = -0.83$); Bottom right: temperature boxplot by season.

Figure 5 — Exposure–outcome scatter plots by season

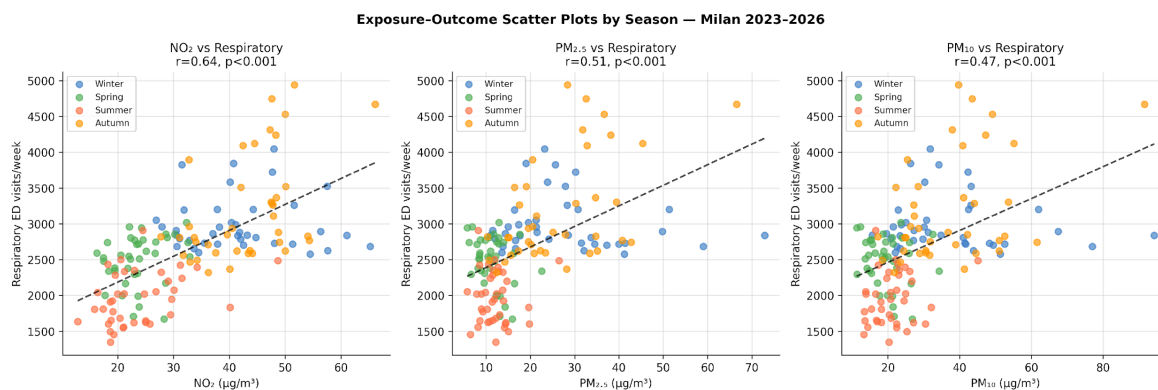


Figure 5. Scatter plots of NO₂, PM_{2.5} and PM₁₀ vs respiratory ED visits, coloured by season. Dashed line = OLS regression. Winter (blue) consistently in upper-right.

Figure 6 — Pollutant seasonal boxplots

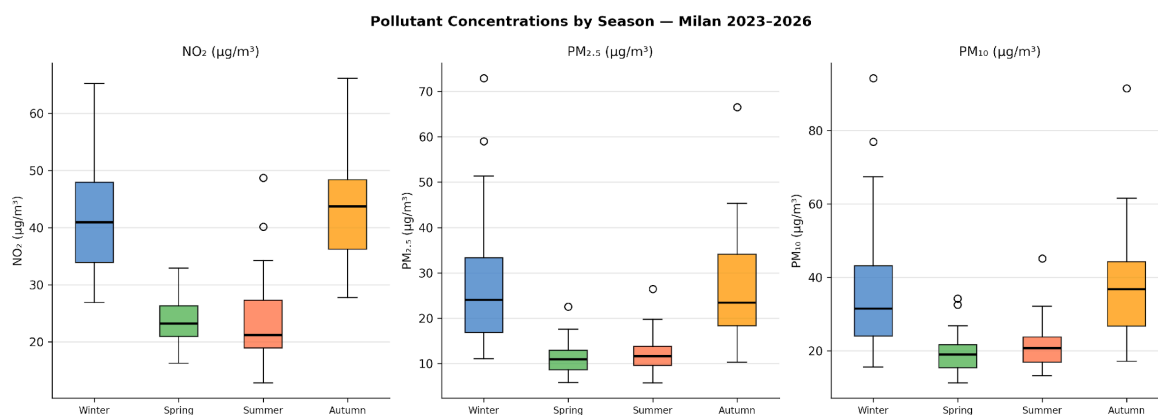


Figure 6. Boxplots of NO₂, PM_{2.5} and PM₁₀ by season. All pollutants significantly higher in winter (Mann-Whitney $p < 0.001$).

Figure 7 — Health outcome seasonal boxplots

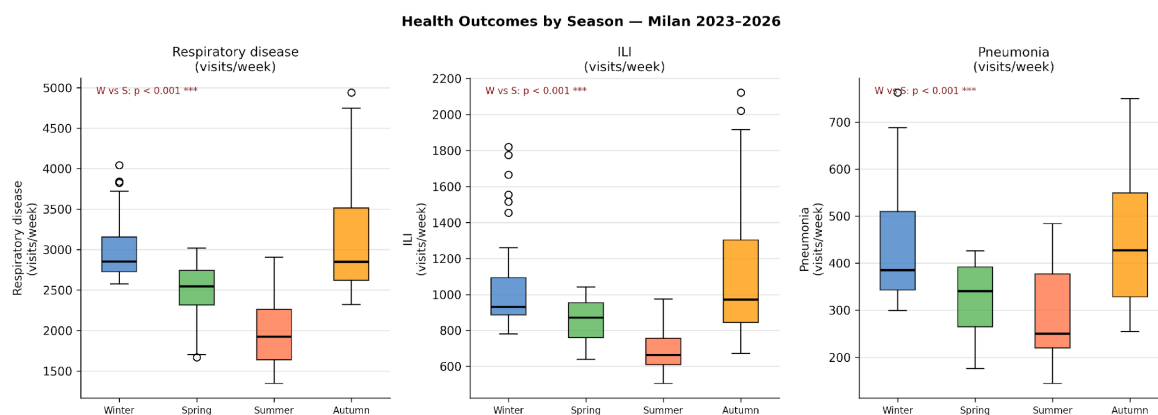


Figure 7. Boxplots of respiratory disease, ILI and pneumonia by season. Winter vs summer: $p < 0.001$ *** for all three outcomes.

Figure 8 — Spearman correlation heatmap

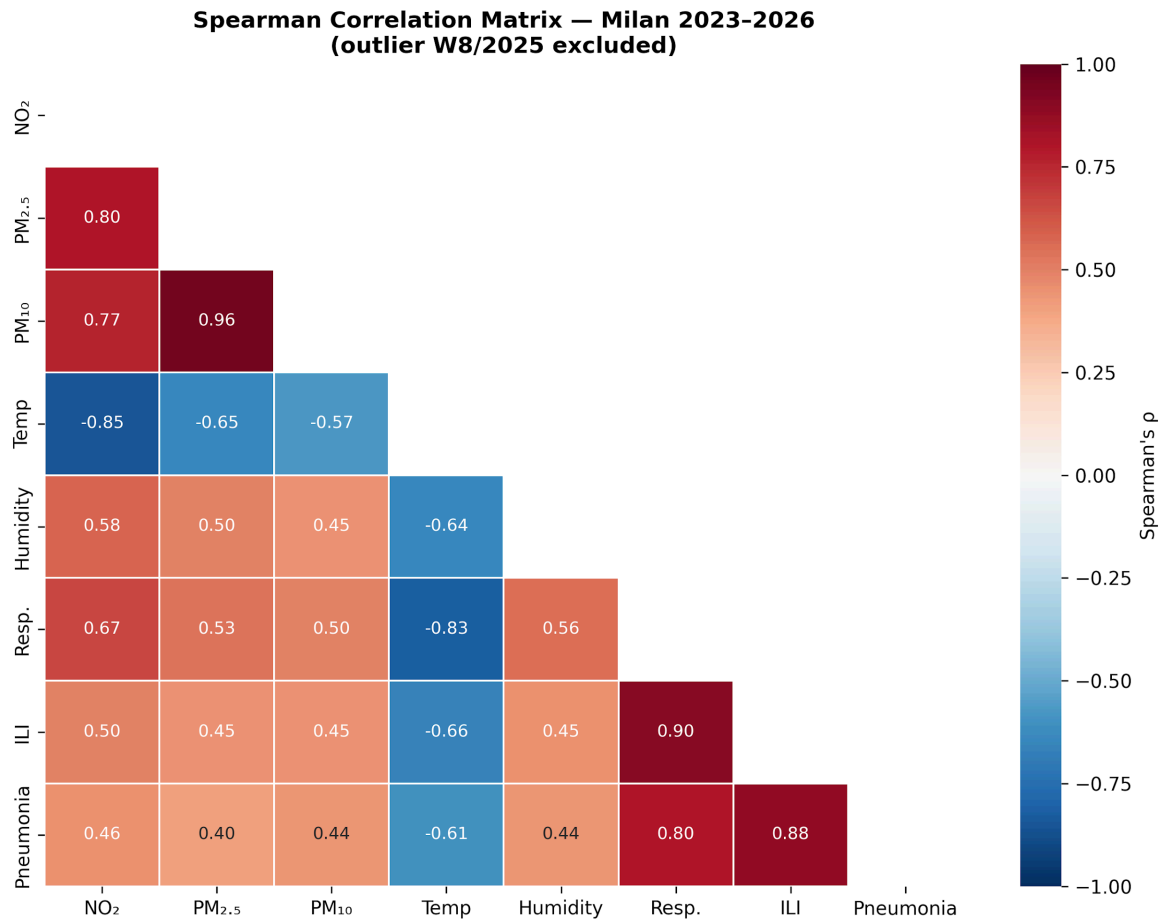


Figure 8. Lower-triangle Spearman correlation heatmap ($n=154$, outlier excluded). Colour scale: blue = negative, red = positive. Notable: $\text{temp} \rightarrow \text{Resp } \rho = -0.83$; $\text{PM}_{2.5} \leftrightarrow \text{PM}_{10} \rho = +0.96$.

Figure 9 — Scatter matrix



Figure 9. Scatter matrix: NO_2 , $\text{PM}_{2.5}$, PM_{10} vs respiratory disease, coloured by season. Diagonal = histograms. Shows near-linear pollutant–pollutant relationships.

Figure 10 — Fourier harmonics verification

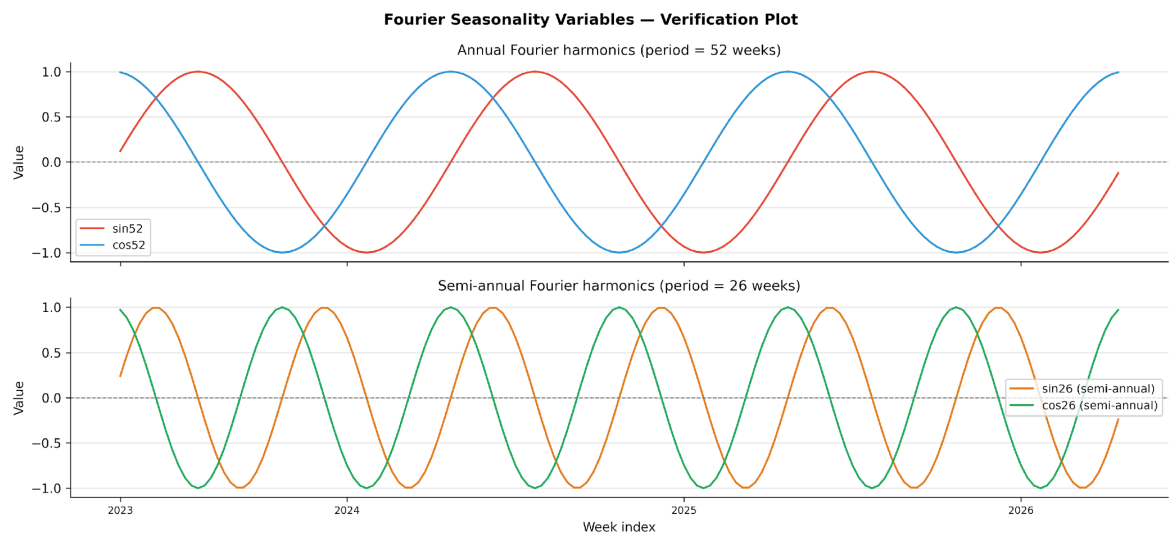


Figure 10. Annual ($\sin 52$, $\cos 52$) and semi-annual ($\sin 26$, $\cos 26$) Fourier harmonics over the full study period. Used by P2 in the DLNM Fourier sensitivity model.